

Qatar-1b

R-0336

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_170409 · created 2026-07-18T05:51:19+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457852.93277 +/- 0.00095 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.011
Transit depth (Rp/Rs)^2	3.68 +/- 0.42 [%]
Orbital Inclination (inc)	83.75 +/- 1.63
Airmass coefficient 1 (a1)	0.944 +/- 0.018
Airmass coefficient 2 (a2)	0.034 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.91 %
Best Comparison Star	#2 - [403, 36]
Optimal Method	PSF photometry
Transit Duration (day)	0.069 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.192 ± 0.011 vs 0.1463 (deviation 3.64σ)
Duration fit vs published	0.069 d vs 0.0692 d (deviation 0.01σ)
Mid-time O-C	2.6 min
Depth SNR	15.3
Residual scatter	1.91 %

Light curve

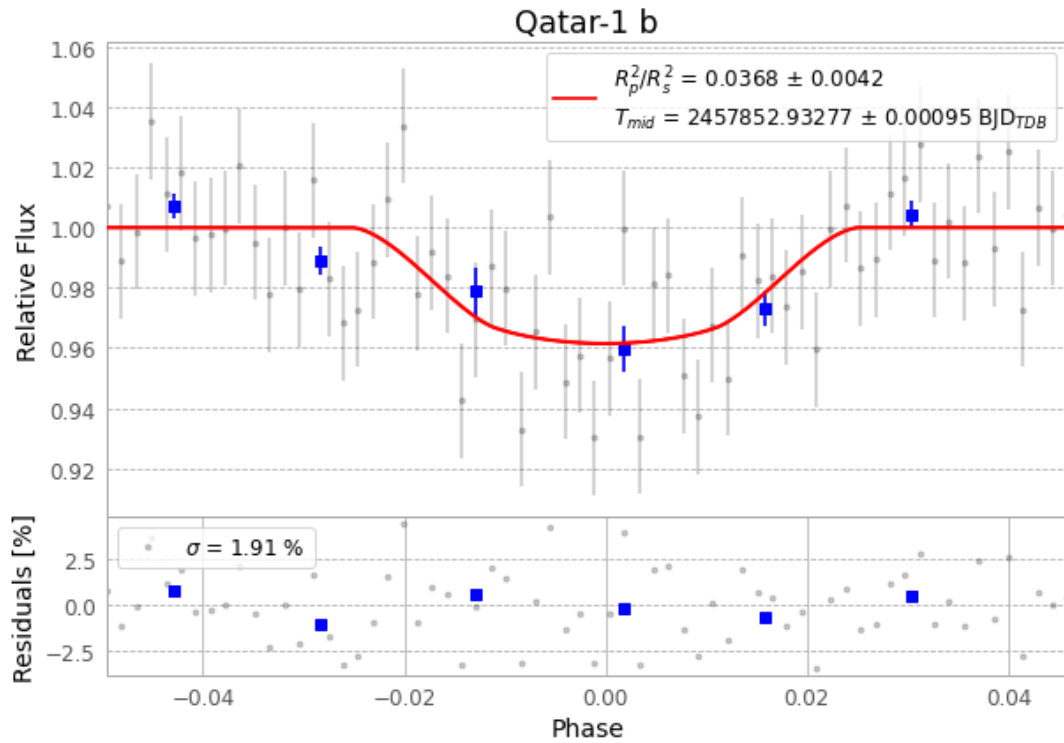


Figure 1 — FinalLightCurve_Qatar-1 b_2017-04-09.png (SHA-256 b011851dd1160448...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [215, 195], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 518434c8542110a0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit d3d5c99

Data provenance

66 FITS frames (43 MB), Qatar-1170409084212.FITS → Qatar-1170409115712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-170409.log (85b9dd772a92...), FinalLightCurve_Qatar-1 b_2017-04-09.png (b011851dd116...), FinalLightCurve_Qatar-1 b_2017-04-09.csv (ad2a70cbe432...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 05:51Z.